

# Tristan Brideweser

**mobile:** (845)-537-5697 | **email:** tristan.brideweser@gmail.com | **github:** <https://github.com/tristanbrideweser> | **portfolio:** <https://tristanbrideweser.github.io>

Dual Master's candidate in Robotics Engineering and Business & Technology with 3+ years of experience leading technical teams in AI, autonomous systems, and cloud infrastructure; bridging complex engineering and business value

## Education

### **Purdue University - Main Campus - West Lafayette, IN**

- **Dual Master of Science, Business - Interdisciplinary Engineering, Business & Technology | May 2027**

- **Concentrations:** Robotics, AI Innovations

- **Bachelor of Science - Data Science, Applied Statistics | May 2025**

- **Minor:** Design & Innovation (Industrial Design)

## Work Experience

- **Lead Technical Researcher | Purdue University - The Data Mine Corporate Partners Program | Aug 2022 - Present**

- **Agile Leadership:** Directed cross-functional teams of student researchers through the full SDLC, facilitating Agile ceremonies and managing stakeholder requirements for four industry-scale projects
- **Johnson & Johnson:** Architected a scalable AI workflow for clinical data harmonization, utilizing automated subgroup identification to accelerate the generation of medical hypotheses.
- **CTB, Inc.:** Developed and deployed predictive maintenance models achieving >85% anomaly detection accuracy for industrial poultry hardware, significantly reducing operational downtime risk.
- **Knudsen Institute:** Engineered a time-series analysis dashboard to cluster procurement trends, enabling supply chain surge forecasting with a 14-day lead time.
- **MISO:** Built high-fidelity regression models to automate transmission line rating forecasts with >80% accuracy, ensuring critical infrastructure compliance with FERC Order 881.

- **Teaching Assistant | Mitchell E. Daniels School of Business - Purdue University | Feb 2024 - May 2024**

- Engineered Python-based integration with Qualtrics API to automate data collection and sentiment analysis for student leadership assessments

- **Artificial Intelligence Research Intern | Weslabs.ai - Boston, MA | May 2023 - Sept 2023**

- Conducted technical market research and competitor benchmarking to define product-market fit for AI-assisted learning platforms, identifying key segments for MVP launch

## Research & Projects

- **Lead Developer | Personal Projects**

- **Horizon:** Architecting an open-source Rust-based terminal user interface (TUI) to streamline ROS2 node management, utilizing asynchronous streams for real-time telemetry visualization.
- **BionicReader:** Developed real-time augmented reality vision pipeline utilizing OCR (Optical Character Recognition) to detect text and dynamically bold it for enhanced readability

- **DevOps Engineer | Fullstack Cloud Marketplace Project | Oct 2025 - Dec 2025**

- Engineered a containerized full-stack architecture deployed on GCP featuring a REST API backend with JWT-based authentication for secure user transactions
- Configured automated deployment pipelines and managed cloud infrastructure

- **Undergraduate Researcher | Purdue UAS Research and Testing (PURT) | Aug 2024 - Dec 2024**

- Enhanced autopilot precision for UAS platforms by implementing Extended Kalman Filters (EKF) in Python, resulting in a 10% reduction in mission flight time.

- **Algorithms Team Lead & Outreach Coordinator | Autonomous Robotics Club of Purdue (ARC) | Jan 2023 - May 2025**

- Directed a 10-member research team in developing C++ implementations of Kalman filters and Viterbi algorithms for real-time state estimation in a robotic piano-playing hand.
- Coordinated partnerships with faculty & industry sponsors, securing funding and resources.

- **Co-Founder & Data Scientist | Startup Project - Spoke & Wing, LLC | Jan 2022 - May 2023**

- Developed predictive models for aviation fuel-tank regulators, estimating \$25-30M in lifetime cost savings by minimizing leakage rates
- Awarded 1st Place in Design & Innovation Challenge 2023 and 2nd Place in Moonshot Pitch Challenge 2024 for technical feasibility and business model scalability

## Skills

**Programming Languages:** Python, C++, Rust, C, SQL, R, Bash

**Robotics & Engineering:** ROS2, OpenCV, Nav2, Sensor Fusion, CAD, PCB Design, Circuit Analysis, Linux

**AI & Data Science:** PyTorch, Jax, Scikit-learn, Time-Series Analysis, Sensor Fusion, Computer Vision

**Cloud & DevOps:** Google Cloud Platform (GCP), Docker, Kubernetes, GitLab CI/CD, Git, Linux, Shell Scripting

**Databases & Data Storage:** SQL (SQLite, MySQL), NoSQL (MongoDB, Redis), Google BigQuery, Database Schema Design

**Management & Methodologies:** Agile/Scrum, Software Development Life Cycle (SDLC), Technical Stakeholder Management

